

Assignment 4 Solutions

5.2

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{dom} : \mathbb{R}^4$$

$$\text{cod} : \mathbb{R}^3$$

5.7

$$\begin{bmatrix} 7 & -5 & -7 & -8 \\ 6 & 1 & 2 & -8 \end{bmatrix}$$

$$\text{dom} : \mathbb{R}^4$$

$$\text{cod} : \mathbb{R}^2$$

5.8

$$T(5\vec{v}_3) = 5 T(\vec{v}_3) = 5 \begin{bmatrix} 8 \\ -7 \\ 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 40 \\ -35 \\ 30 \\ 30 \end{bmatrix}$$

5.11

$$T(2\vec{v}_1 + 2\vec{v}_2 - 4\vec{v}_3) =$$

$$2T(\vec{v}_1) + 2T(\vec{v}_2) - 4T(\vec{v}_3) =$$

$$2 \begin{bmatrix} -6 \\ 3 \\ -2 \\ -10 \end{bmatrix} + 2 \begin{bmatrix} -5 \\ 1 \\ -2 \\ 9 \end{bmatrix} - 4 \begin{bmatrix} 8 \\ -7 \\ 6 \\ 6 \end{bmatrix} =$$

$$\begin{bmatrix} -12 & -10 & -32 \\ 6 & 2 & 28 \\ -4 & -4 & -24 \\ -20 & 18 & -24 \end{bmatrix} = \begin{bmatrix} -54 \\ 36 \\ -32 \\ -26 \end{bmatrix}$$

$$5.12 \quad \begin{bmatrix} 3 & -3 & 21 \\ 2 & 1 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 7 \\ 2 & 1 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 7 \\ 0 & 3 & -14 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 7 \\ 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & -2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 5 \\ -2 \end{bmatrix}\right) = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$$

Unique because there are no free variables in solution to $A\vec{x} = \vec{v}$

$$5.14 \quad \begin{bmatrix} 2 & 6 & 6 & -30 & 0 \\ 2 & 6 & 3 & -21 & 9 \\ -1 & -3 & -1 & 9 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -15 & 0 \\ 2 & 6 & 3 & -21 & 9 \\ -1 & -3 & -1 & 9 & -2 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & 3 & 3 & -15 & 0 \\ 0 & 0 & -3 & 9 & 9 \\ 0 & 0 & 2 & -6 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 & -15 & 0 \\ 0 & 0 & 1 & -3 & -3 \\ 0 & 0 & 1 & -3 & -1 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 3 & 3 & -15 & 0 \\ 0 & 0 & 1 & -3 & -3 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$v \notin \text{ran}(T)$

5.17

$$\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = T\left(-3\begin{bmatrix} 1 \\ -2 \end{bmatrix} + 2\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$$

$$= -3 T\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right) + 2 T\left(\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$$

$$= -3 \begin{bmatrix} -3 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -4 \\ -1 \end{bmatrix} = \begin{bmatrix} 9 & -8 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ -2 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = T\left(-2\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$$

$$= -2 T\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right) + T\left(\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right)$$

$$= -2 \begin{bmatrix} -3 \\ -1 \end{bmatrix} + \begin{bmatrix} -4 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \vec{x}$$

5.20

$$\begin{aligned}
 \begin{bmatrix} -7 & 9 & 1 \\ -3 & 4 & 0 \end{bmatrix} &\sim \begin{bmatrix} 21 & -27 & -3 \\ -21 & 28 & 0 \end{bmatrix} \sim \begin{bmatrix} -7 & 9 & 1 \\ 0 & 1 & -3 \end{bmatrix} \\
 &\sim \begin{bmatrix} -7 & 0 & 28 \\ 0 & 1 & -3 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -3 \end{bmatrix}
 \end{aligned}$$

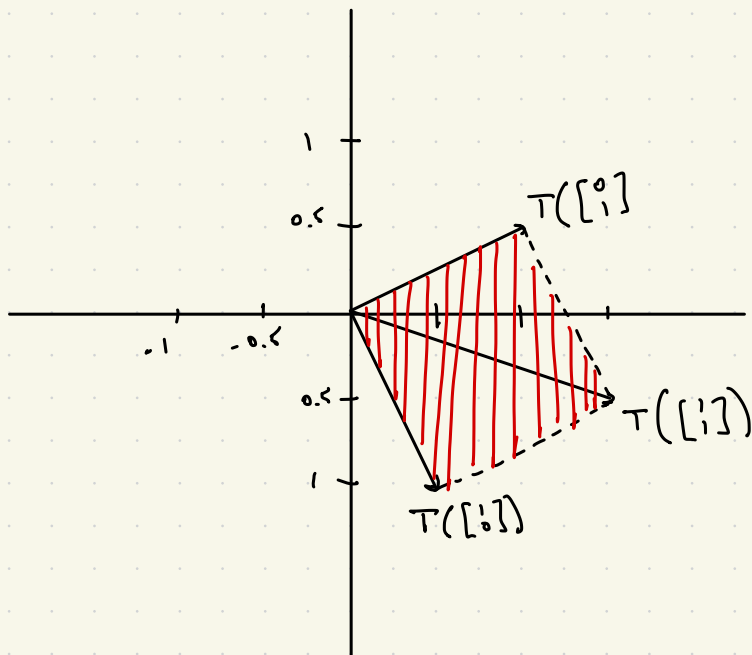
$$T\left(-4 \begin{bmatrix} -7 \\ -3 \end{bmatrix} - 3 \begin{bmatrix} 9 \\ 4 \end{bmatrix}\right) =$$

$$-4 \begin{bmatrix} -4 \\ 3 \\ 4 \\ 0 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 13 \\ 0 \\ 0 \\ 8 \\ 3 \end{bmatrix} = \begin{bmatrix} 16 - 39 \\ -12 \\ -16 \\ -24 \\ -4 - 9 \end{bmatrix} = \begin{bmatrix} -23 \\ -12 \\ -16 \\ -24 \\ -13 \end{bmatrix}$$

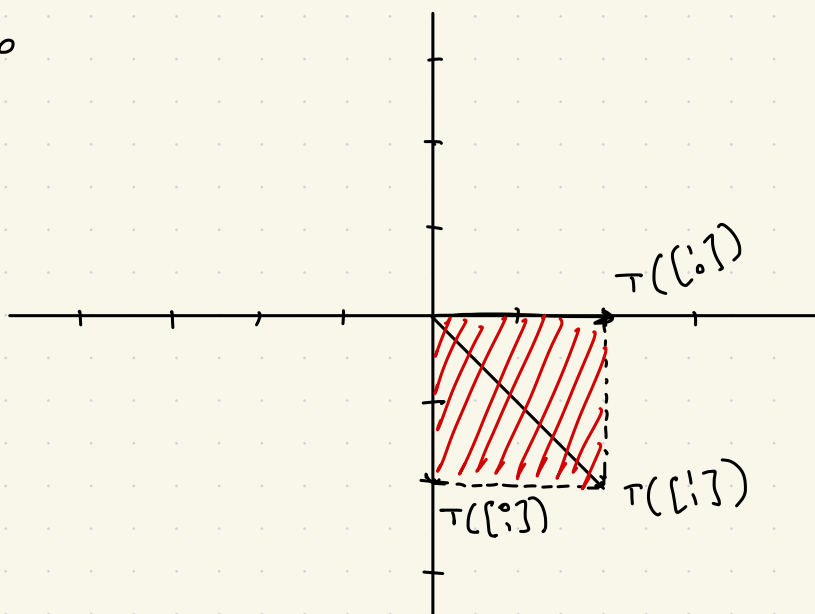
$$\begin{aligned}
 T\left(\begin{bmatrix} 0 \\ 4 \end{bmatrix}\right) &= T\left(\left(\begin{bmatrix} 9 \\ 4 \end{bmatrix} - 9 \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)\right) \\
 &= \left(T\left(\begin{bmatrix} 9 \\ 4 \end{bmatrix}\right) - 9 T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)\right) \\
 &= \begin{bmatrix} 13 \\ 0 \\ 0 \\ 8 \\ 3 \end{bmatrix} - 9 \begin{bmatrix} -23 \\ -12 \\ -16 \\ -24 \\ -13 \end{bmatrix} = \begin{bmatrix} 55 \\ 27 \\ 36 \\ 56 \\ 30 \end{bmatrix}
 \end{aligned}$$

$$\begin{bmatrix} -23 & 55 \\ -12 & 27 \\ -16 & 36 \\ -24 & 56 \\ 13 & 30 \end{bmatrix}$$

5.35



5.36



1.44 True

$$\begin{bmatrix} 0 & 0 & \dots & 0 & 0 & \blacksquare \end{bmatrix}$$

leading entry

1.46 False

$$\begin{array}{c} R_1 \leftrightarrow R_2 \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_1 - 2R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{array}$$

1.52 True Same reasoning as 1.44

3.18 False

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

3.24 False

$A =$ all zeros matrix

$$\vec{b} = \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

4.8 True

entries above and below pivots in every column are 0

4.11 True

$$\begin{array}{l} a_1 \vec{v}_1 + a_2 \vec{v}_2 = \vec{0} \quad (a_1, a_2) \text{ nontrivial} \Rightarrow \\ a_1 \vec{v}_1 + a_2 \vec{v}_2 + 0 \vec{v}_3 = \vec{0} \quad (a_1, a_2, 0) \text{ nontrivial} \end{array}$$

5.41 False

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

5.42 False should be $n \times m$.

5.43 True $T(\vec{e}_i)$ is uniquely determined

5.46

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

5.47

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

5.48

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

5.49

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

5.52

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ 2x \end{bmatrix}$$

$$-\alpha = 2x \Rightarrow \alpha = -2x$$

$$1 + 2 = x \Rightarrow$$

$$1 - 4x = x \Rightarrow$$

$$x = \frac{1}{5} \quad \alpha = -\frac{2}{5}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \frac{4}{5} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$$

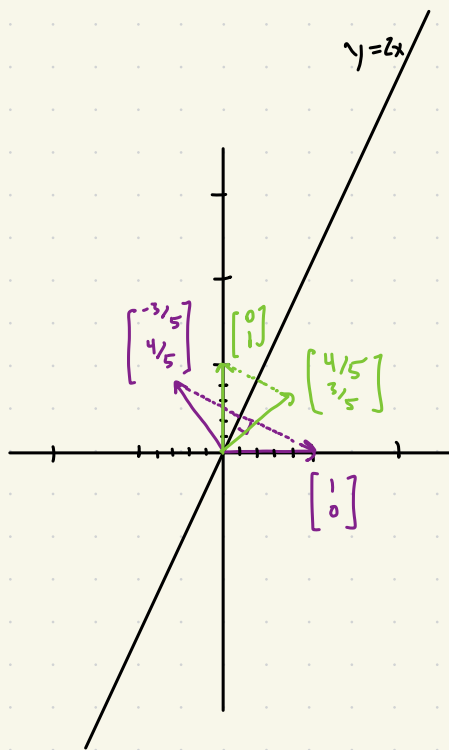
$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \alpha \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} x \\ 2x \end{bmatrix}$$

$$2\alpha = x \quad \alpha = \frac{x}{2}$$

$$1 + \frac{-x}{2} = 2x$$

$$1 = \frac{5}{2}x \quad x = \frac{2}{5} \quad \alpha = \frac{1}{5}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} + \frac{2}{5} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}$$



$$\begin{bmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{bmatrix}$$

5.53

$$\begin{aligned}
 a) \quad \begin{bmatrix} cx \\ cy \\ cz \end{bmatrix} &= \begin{bmatrix} ((cx)^3 + (cy)^3 + (cz)^3)^{1/3} \\ cy + cz \end{bmatrix} = \begin{bmatrix} (c^3(x^3 + y^3 + z^3))^{1/3} \\ c(y+z) \end{bmatrix} \\
 &= \begin{bmatrix} c(x^3 + y^3 + z^3)^{1/3} \\ c(y+z) \end{bmatrix} = c \begin{bmatrix} (x^3 + y^3 + z^3)^{1/3} \\ y+z \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &\mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} &\mapsto \begin{bmatrix} \sqrt[3]{2} \\ 1 \end{bmatrix} \\
 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} &\mapsto \begin{bmatrix} 1 \\ 1 \end{bmatrix}
 \end{aligned}$$

$$\tau\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) + \tau\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \neq \begin{bmatrix} \sqrt[3]{2} \\ 1 \end{bmatrix} = \tau\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right)$$